

1. Why is Python called a Dynamically Typed Language?

Python is dynamically typed because you do not need to declare the data type of a variable. Python automatically determines the type based on the assigned value.

2. Why is Python Platform Independent?

Python programs can run on different operating systems such as Windows, Linux, and macOS without changing the source code.

Write Once, Run Anywhere (with a compatible Python interpreter).

3. Explain the Python Execution Process.

Source Code (.py)



Python Compiler



Bytecode (.pyc)



Python Virtual Machine (PVM)



Output

4. What is Bytecode?

Bytecode is an intermediate code generated after compiling Python source code. It is stored as .pyc files and executed by the Python Virtual Machine.

5. What is Python Virtual Machine (PVM)?

The Python Virtual Machine (PVM) reads the bytecode and executes it line by line, producing the program's output.

